PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Detail
Pentester Name (Cybersecurity Professional)	**Razak Moro**
Program/Batch	B084-Networkwalks
Date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (theHarvester) W2-PM5 (Network Scanning with Nmap/Zenmap)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local lab network (VirtualBox)
Permission secured from client?	Yes (Letter of Authorization, ref. Networkwalks)
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

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1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the `networkwalks.com` domain using multiple Kali Linux tools (W2-PM1), running theHarvester (W2-PM4) and scanning my own local lab network with Nmap (W2-PM5). The footprinting modules cover the information-gathering phase and the Nmap module covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux running in Oracle VirtualBox. Every step below includes the exact command used, the result I observed, a screenshot as evidence (Section 8), and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

Tool	Purpose
Kali Linux (VirtualBox)	Operating system used for all reconnaissance and scanning activities
WHOIS	Find domain registration details (registrar, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, TXT/SPF, SRV).

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Tool	Purpose
Nmap (`-sn` ping scan)	Scan the local subnet to find live hosts, IPs and MAC addresses. This is the command-line engine behind Zenmap's "Ping scan" profile.
theHarvester	Collect emails, hosts and IPs for a domain from public search-engine sources.
ifconfig	Identify the local IP address and subnet.

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against `networkwalks.com` using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool collected a different type of information about the target. Only passive and light information-gathering was done; nothing was exploited.

1. WHOIS — `whois networkwalks.com`
The domain is registered with **GoDaddy.com, LLC**. It was created on **6 November 2019** and is set to expire on **6 November 2027**. The name servers are **NS6135.HOSTGATOR.COM** and **NS6136.HOSTGATOR.COM**, and **DNSSEC is unsigned**. The registrar record shows the registrant details are hidden behind a privacy service (Domains By Proxy, LLC).
Why it matters: Registration data and name servers tell an attacker who hosts the domain, and where to look next.

2. WhatWeb — `whatweb networkwalks.com`
The site redirects HTTP to HTTPS (301) and runs on **Apache**. It identified **WordPress 7.1.4** with the **WordPress Download Manager 3.3.76** plugin, **jQuery 3.7.1**, the page title "Networkwalks Academy", the public contact email `info@networkwalks.com`, and the IP **192.232.216.135** (United States). The `__wpdm_client` cookie is flagged HttpOnly.
Why it matters: A named CMS and plugin version lets an attacker search for known issues affecting that exact software.

3. Nslookup — `nslookup networkwalks.com`
The domain resolved to **192.232.216.135** via the 8.8.8.8 resolver (non-authoritative answer). The first query printed a timeout message before the answer was returned.
Why it matters: It confirms the public IP that later scanning would target.

4. Curl — `curl -I https://networkwalks.com`
The server replied **HTTP/2 200** with `server: Apache`. The headers expose the WordPress REST API endpoint (`https://networkwalks.com/wp-json/`) and a page reference (`/wp-json/wp/v2/pages/53`), caching headers (`x-nginx-cache: WordPress`, `x-endurance-cache-level`), a `referrer-policy` of `no-referrer-when-downgrade`, and a `permissions-policy`. The `__wpdm_client` cookie is set with `secure; HttpOnly`. Common hardening headers such as `Strict-Transport-Security`, `Content-Security-Policy`, `X-Frame-Options` and `X-Content-Type-Options` were **not present** in the captured response.
Why it matters: Headers confirm the technology stack and reveal which browser-side protections are missing.

5. Wafw00f — `wafw00f networkwalks.com`
Wafw00f v2.3.1 reported that the site is behind **ModSecurity (SpiderLabs)** WAF, using 2 requests.
Why it matters: An attacker would know a WAF is filtering requests and would adjust their approach. For the owner, this is a positive control.

6. DNSRecon — `dnsrecon -d networkwalks.com`

- **DNSSEC** is not configured.
- **SOA / NS:** `ns6135.hostgator.com` (50.87.144.87) and `ns6136.hostgator.com` (192.232.216.131).
- **BIND version** disclosed by the name server at 50.87.144.87: `9.16.23-RH`.
- **A / MX:** `networkwalks.com` and `mail.networkwalks.com` both resolve to 192.232.216.135.
- **TXT:** an SPF record (`v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`) and a Google site-verification record.

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- **SRV:** 8 `_autodiscover._tcp` records pointing to `cpanelemaildiscovery.cpanel.net`.

I ran the command twice. The second run returned the SRV records that the first run did not, most likely because of DNS timeouts, so the results above combine both runs.
Why it matters: DNS records map the hosting provider, mail setup and software in use, which helps an attacker build an infrastructure profile.

4.2 theHarvester (W2-PM4)

I ran theHarvester 4.8.0 with the Baidu search engine as the data source:

```
theHarvester -d microsoft.com -l 1000 -b baidu
```

The tool printed repeated "An exception has occurred" messages and then reported **no IPs, no emails, no people and no hosts found** for `microsoft.com`.

Scope note: `microsoft.com` is not one of the targets named in my Letter of Authorization (`networkwalks.com` and my own network). The tool queried Baidu's search results rather than Microsoft's systems, and it returned no data, but this run was outside the written scope. It should be repeated against `networkwalks.com`, which is in scope.
Why it matters: Tools like theHarvester show how much of an organization's emails and subdomains can be collected from public sources without touching its servers.

4.3 Network Scanning with Nmap (W2-PM5)

For the second activity, I performed host discovery on my own lab network. I first used `ifconfig` to identify my local IP address and subnet: the Kali machine's interface `eth0` has the address **10.0.0.2** with netmask **255.255.255.0**, so the subnet is **10.0.0.0/24**. I then ran a ping scan, the same scan that Zenmap runs with its **Ping scan** profile:

```
nmap -sn 10.0.0.0/24
```

Nmap 7.95 scanned 256 IP addresses and found **5 hosts up** in 30.55 seconds, on 10 November 2026:

IP Address	MAC Address	Notes
10.0.0.1	52:55:0A:00:00:01	Vendor shown as "Unknown"
10.0.0.3	52:55:0A:00:00:03	Vendor shown as "Unknown"
10.0.0.4	08:00:27:E3:AB:75	Oracle VirtualBox virtual NIC
10.0.0.5	08:00:27:77:FF:E9	Oracle VirtualBox virtual NIC
10.0.0.2	08:00:27:1A:AE:52	My own Kali VM (the scanning machine; Nmap does not show its own MAC)

Why it matters: Host discovery is how an attacker maps which devices are alive before choosing targets for deeper scanning.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology and version information exposed	WhatWeb identified WordPress 7.1.4 and WordPress Download Manager 3.3.76	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	DNS server software version disclosed	DNSRecon returned BIND version `9.16.23-RH` from 50.87.144.87	Helps an attacker target known issues for that DNS software. The `-RH` suffix suggests a vendor build that may carry back-ported patches, so this needs verification	● Medium

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#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
3	DNSSEC not configured	WHOIS shows DNSSEC unsigned; DNSRecon confirmed it is not configured	DNS answers cannot be cryptographically validated, which leaves room for DNS spoofing attacks	● Medium
4	Server IP address identifiable	Nslookup resolved the domain to `192.232.216.135`	Provides the network location of the web service	● Low
5	HTTP technical information exposed and hardening headers absent	Curl showed `/wp-json/` and the page reference in headers; no HSTS, CSP, X-Frame-Options or X-Content-Type-Options in the response	May assist technology fingerprinting and leaves browser-side protections unused	● Low
6	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
7	Public contact email and mail/SPF details visible	WhatWeb found `info@networkwalks.com`; DNSRecon showed MX, SPF (`~all`) and SRV records	Could be used for targeted phishing or mail-spoofing attempts	● Low
8	Multiple live hosts visible on the local network	Nmap found 5 live hosts on 10.0.0.0/24	Unknown or unauthorized devices could be present on a network. All five hosts here are part of my own lab setup	● Low

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
   WordPress, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   Remove unnecessary technical headers and add standard security headers such as HSTS, Content-Security-Policy, X-Frame-Options and X-Content-Type-Options.
4. **Review DNS records and DNS software**
   DNS records should be checked periodically so only required information is public. Hide the DNS server version string and confirm the name servers are patched.
5. **Consider enabling DNSSEC**
   Enabling DNSSEC with the DNS provider would allow resolvers to validate the domain's DNS answers.
6. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
7. **Tighten the SPF policy**
   Once all legitimate mail sources are confirmed, consider moving SPF from `~all` (soft fail) to `-all` (hard fail), and add DMARC if it is not already in place.

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8. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
9. **Investigate unknown devices and document the network**
   Any unexpected device found during scanning should be verified, and network topology and device information should be kept up to date.
10. **Perform security testing with authorization**
    Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain registration information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can enumerate DNS records and reveal software details.

In the network scanning activity, I used Nmap to identify my local network configuration and discover active hosts on my lab network. I collected the IP and MAC address of each host.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. The footprinting and scanning activities were completed against the in-scope targets (`networkwalks.com` and my own network) as part of the assigned educational cybersecurity lab. The one exception is the theHarvester run against `microsoft.com` (Section 4.2), which was outside my written scope and should be repeated on an in-scope target.

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8. Evidences Collected

1. WHOIS

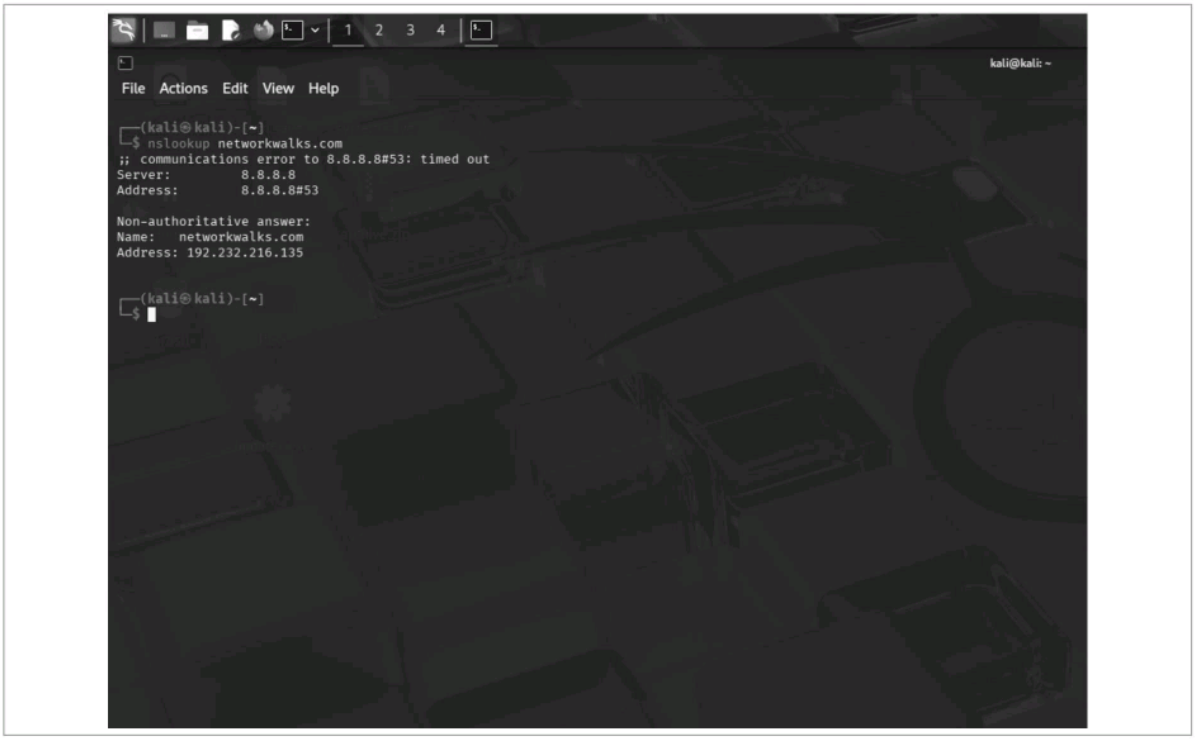


2. WhatWeb



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3. Nslookup



4. Curl



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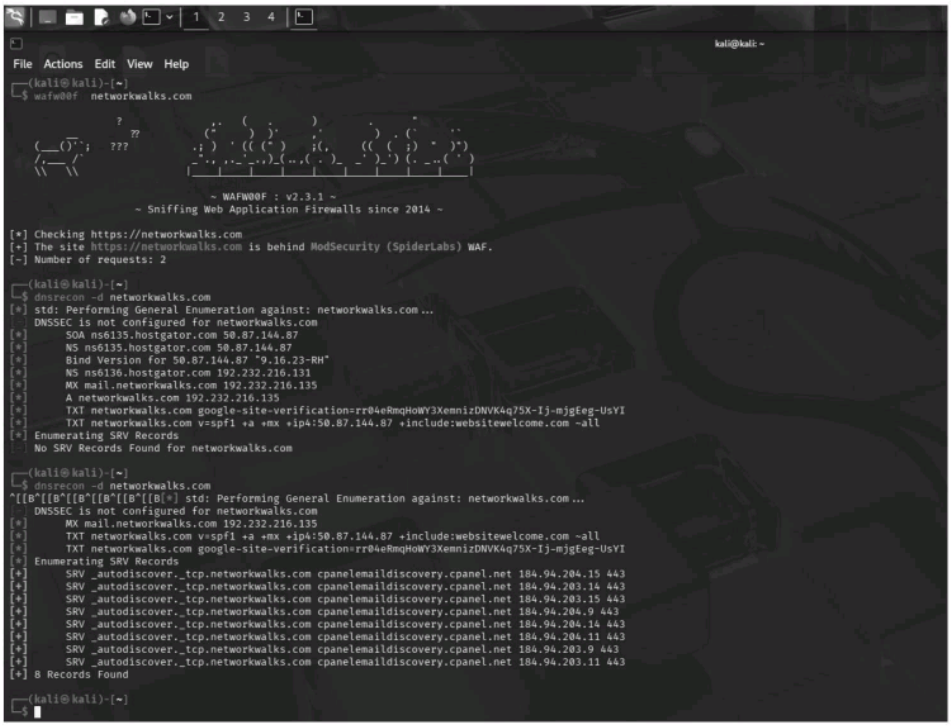
5. Wafw00f

```
┌──(kali㉿kali)-[~]
└─$ wafw00f  networkwalks.com

                ~ WAFW00F : v2.3.1 ~
          ~ Sniffing Web Application Firewalls since 2014 ~

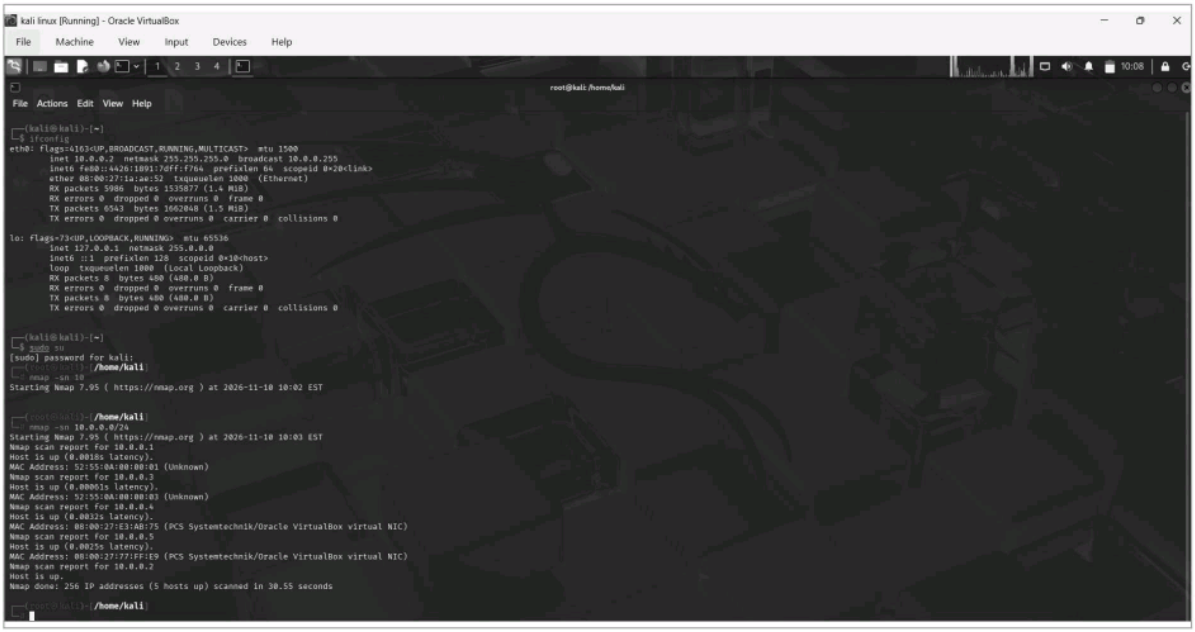
*] Checking https://networkwalks.com
+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
~] Number of requests: 2
```

6. DNSRecon

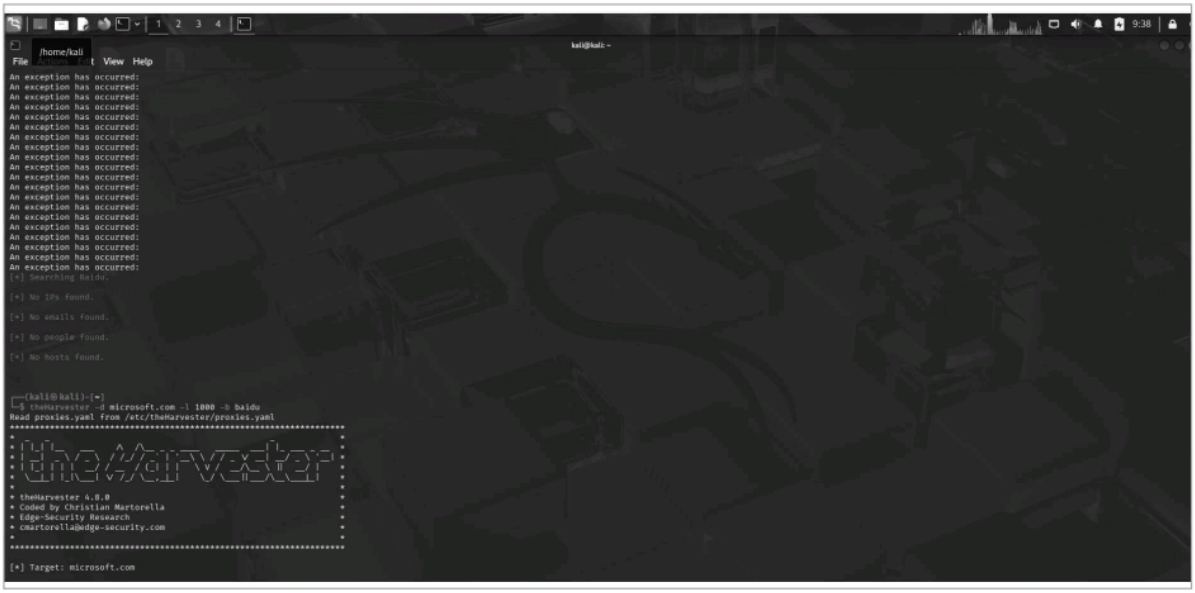


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7. Nmap ping scan of the local network (including ifconfig)



8. theHarvester (microsoft.com, Baidu source)



-End-

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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub

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